CSA15 Cloud Computing and Big Data Analytics

Cloud Security

Capstone Cloud Technical Presentation — Individual Student Presentation Deck

Presentation Focus

Presenting Cloud Security as a cloud-backed product: the problem, architecture, service choices, evidence, and CO1 learning behind protecting identity, data, and access in the cloud.

Presenter Details

Student Name:

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Register Number:

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Class / Section / Slot:

CSA 1512 slot A

Capstone Title:

Cloud Security

Selected SDG Goal(s):

SDG 9 – Industry, Innovation & Infrastructure*

Cloud Platform / Tools:

AWS – IAM, KMS, CloudTrail, GuardDuty*

**Placeholder — update Class/Section/Slot, SDG goal, and platform to match your actual work before submission.*

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

Organizations moving workloads to the cloud face rising risk from misconfigured access, weak identity controls, and undetected intrusions — leading to data breaches, downtime, and compliance failures.

Target Users

Cloud administrators and security teams who need real-time visibility into access and threats, and end-users whose personal data is stored in the cloud and must stay protected.

SDG Fit

SDG 9 – Industry, Innovation and Infrastructure: the project strengthens resilient, trustworthy digital infrastructure that organizations can safely build on.

Product Boundary

First usable version: authenticated login with MFA, role-based access control, encrypted storage, and a monitoring dashboard that flags suspicious activity.

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

Cloud Definition

For Cloud Security, 'cloud' means using a provider's on-demand identity, storage, and monitoring services rather than building and hardening security infrastructure in-house.

Cloud Characteristics

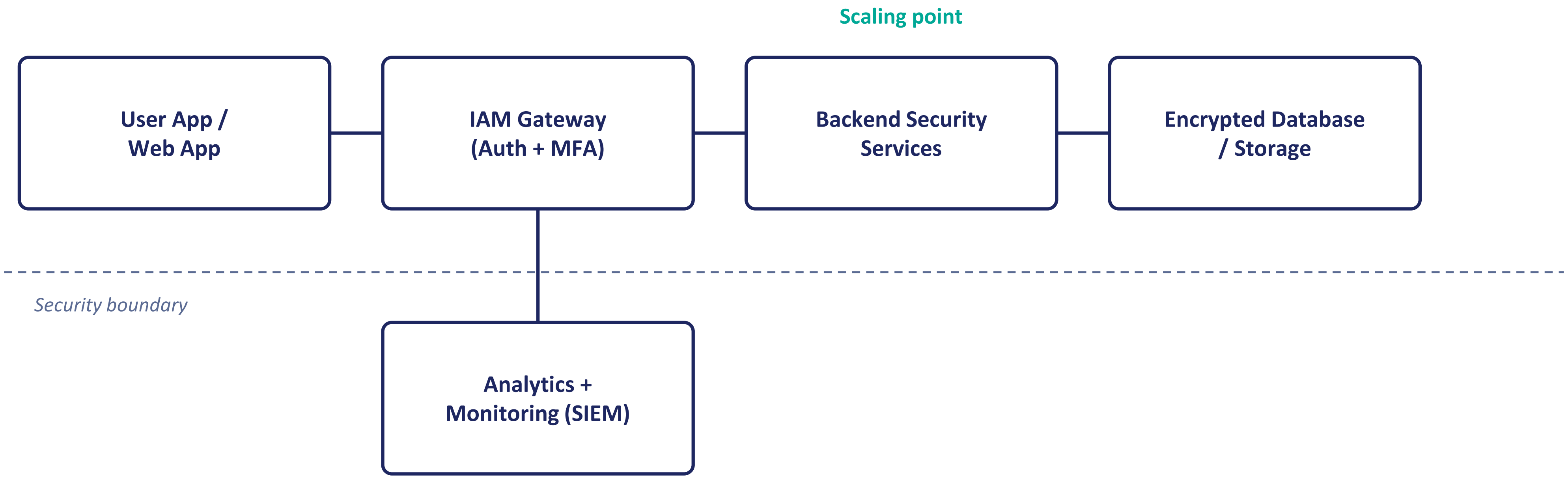
Elasticity absorbs spikes in login/authentication requests; scalability grows storage as logs accumulate; measured service means paying only for security tooling used; resource pooling shares the provider's hardened security services across tenants.

Cloud Concept Map

Cloud Ecosystem	IAM · KMS · GuardDuty · CloudTrail
Service Models	IaaS / PaaS / SaaS Security Layers
Elasticity	Auto-scaled Auth & Log Ingestion
Capstone Modules	Auth · Access Control · Monitoring

Capstone Cloud Architecture

Our cloud architecture connects the user action to secure services



Presentation Points

- Request path
- API role
- Compute choice
- Database/storage
- Monitoring & security
- Scaling point

Cloud Service Model Responsibility Matrix

Each service model defines what we control and what the cloud manages

Service area	Model	Provider manages	Student team manages	Why this fits
Frontend hosting	SaaS / PaaS	Hosting, CDN, patching	UI code, security headers	Reduces attack surface at the hosting layer
Backend / API	PaaS	Runtime, OS patching	IAM policies, API logic	Keeps focus on security logic, not infra
Database	Managed DBaaS	Backups, replication, at-rest encryption	Schema, access policy, key rotation	Protects data without managing servers
Storage	Managed object storage	Durability, infrastructure	Bucket policies, encryption keys	Prevents public-exposure misconfiguration
Monitoring / logs	Managed monitoring (SaaS)	Log infrastructure, retention	Alert rules, dashboards	Enables real-time threat detection
Security / IAM	Managed IAM (PaaS)	Identity infra, MFA delivery	Roles, policies, least privilege	Central to enforcing zero-trust access

Data Flow and Analytics Pipeline

Security data moves from user activity to a defensible insight

Collect — **Ingest** — **Store** — **Process** — **Visualize**

Data Requirements

Login attempts, access-control decisions, API request logs, and file-access events across the system.

Analytics Requirement

A live dashboard and alerting layer for anomaly detection, access-pattern trends, and incident reporting.

Proof Required

A sample access log, one flagged anomaly, and the resulting insight — e.g. a spike in failed logins from an unusual location.

Python / AI Module Integration

Python adds intelligence where the security product needs it

Python Role

Anomaly detection over access logs using rule-based checks and statistical scoring to flag risky sessions.

Input and Output

Input: a stream of access-log entries (user, timestamp, location, action). Output: a risk score and a flag on suspicious sessions.

Execution Point

A serverless function triggered whenever a new log entry lands in the pipeline, keeping detection near real time.

Product Value

Cuts manual log review time and surfaces high-risk events to security admins faster than periodic manual audits.

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

Multi-factor authentication, role-based access control, input validation on every endpoint, and a full audit trail of admin actions.

Privacy

Personally identifiable data is encrypted at rest and in transit; consent is captured at signup; log access is restricted by role.

Scalability

Managed identity, storage, and log services scale automatically as users and log volume grow, without redesigning the architecture.

Cost Awareness

Development stays within free-tier compute and storage limits; API and storage usage are monitored to avoid unexpected charges.

Implementation Stack and Deployment Evidence

The implementation stack shows how the design will be built

Web Layer

A React-based web dashboard used by security admins to review alerts, sessions, and access logs.

Cloud Services

Managed compute, managed database, object storage, managed IAM/authentication, log analytics, and an API gateway.

Evidence

GitHub repository, API test results, a cloud monitoring dashboard screenshot, and the deployed endpoint URL.

Version Control

An organized repository with feature branches, descriptive commits, and a README describing setup and architecture.

Demo Storyboard and User Journey

A complete user journey proves the security product flow

Scene 1

User attempts login → system challenges with MFA → IAM gateway verifies identity

Scene 2

User opens a file → system checks role permission → IAM/authorization service validates access

Scene 3

Unauthorized attempt occurs → system blocks and logs it → monitoring service records the event

Scene 4

Admin opens dashboard → system shows recent alerts → analytics service aggregates the logs

Scene 5

Anomaly appears in login pattern → system raises an alert → function scores and flags the session

Scene 6

Admin reviews and resolves the alert → system updates the audit trail → storage records resolution

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

- ☐ Cloud ecosystem and service model explained
- ☐ Elasticity and scalability decisions are visible
- ☐ REST/SOAP/API role connected to capstone
- ☐ GitHub and cloud evidence are inspectable
- ☐ Architecture has compute, data, storage, security, monitoring
- ☐ References and integrity statement are included

Evaluator Focus

CO1 is verified through cloud design reasoning, service choices, evidence, and individual explanation.

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

The project clarified how identity and access management, encryption, and monitoring work together under the shared responsibility model.

Limitations

Anomaly detection currently relies on rule-based thresholds rather than a trained ML model; multi-region resilience is not yet implemented.

Future Improvement

Add a trained anomaly-detection model and extend monitoring to multi-region deployments before the final capstone review.

Integrity Statement

Cloud provider documentation and AI assistance (Claude) supported drafting and structuring; all architecture decisions were reviewed by the student.